

Sarthak Jain

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EDUCATION

Carnegie Mellon University / *B.S. Electrical & Computer Engineering and Robotics* / **GPA 3.8/4.0** Expected May 2028

- **Coursework:** Signals & Systems, Structure & Design of Digital Systems, Introduction to Computer Systems, Embedded Systems, Computer Vision, Fundamentals of Controls, Electronics & Circuits, Principles of Imperative Computation, Linear Algebra

EXPERIENCE

AirLab, CMU Robotics Institute / *Robotics Autonomy Researcher* May 2026 – Present

- Enabled a drone to sense nearby walls with **zero added sensors** (no camera or LiDAR), reaching **under 3 cm** distance and **under 4°** angle error, by training a **physics-informed PyTorch** model on its own thrust, RPM, and IMU signals.
- Produced a **120,000-sample** labeled training set without risking hardware, by building an **ANSYS Fluent CFD digital twin** of near-surface airflow and closing the sim-to-real gap with domain randomization.

Teaching Assistant, 18-290 Signals and Systems / *Carnegie Mellon University, ECE* Fall 2026

- Supported **100+ students**, by teaching recitations and office hours on Fourier/Laplace transforms, FFT, sampling, and filters.
- Assessed student mastery of signal processing each week, by writing and grading problem sets and exams.

ForAll Lab, Carnegie Mellon / *Neuroscience / Signals Engineer* Summer 2026

- Aligned brain-implant recordings from separate machines to **within 2 ms**, a **~30×** reduction in timing spread, by modeling clock drift with least-squares fitting in **NumPy/SciPy**.
- Recovered **100% of event markers**, by de-clustering noisy trigger bursts and adding automated data validation.

Kantor Lab, Carnegie Mellon / *Research Assistant* 2025 – Present

- Helped **win the Farm Robotics Challenge (\$25K)**, by deploying real-time **YOLOv8** inference that detects and grades apple-tree disease on an autonomous robot.
- Automated marking of infected branches, by building the **ROS 2** pipeline linking detections to robotic-arm motion.

IEEE x CMU / *Electrical Lead, animatronic control system* Mar 2025 – Present

- Delivered a real-time animatronic controller leading a **6-person** team, by architecting its **RTOS task system in embedded C**.

Fleeca / *Engineering Intern* Jan 2023 – Jul 2023

- Raised production throughput **22%**, by designing an automated wire cutting/stripping machine and custom fixtures.

PROJECTS

Edge-AI Health Vest | *TinyML, INT8 quantization, sensor fusion, Kalman filtering* 2024 – 2025

- Achieved **under 8 ms** inference on a **sub-milliwatt** power budget with no phone or cloud, by **pruning and INT8-quantizing** the health model to run on the vest's microcontroller.
- Cut motion noise **43%** on a sprinting athlete, by fusing **7 biosignal channels** (heart rate, SpO2, HRV, temperature, IMU) with a denoising autoencoder and adaptive **Kalman filter**.
- Put the vest in the hands of **500+ blind football players**, by partnering with Scout Me to produce and distribute it.

Pipelined RISC-V Processor | *SystemVerilog, computer architecture, Vivado, QuestaSim* 2025

- Kept **5 instructions in flight** at once with **0 hazard failures**, by designing hazard-detection and forwarding logic for an RV32I CPU in SystemVerilog.
- Verified **100% cycle-accurate** behavior, by building a QuestaSim testbench and pipeline visualizer; synthesized in Vivado.

Shadow Fight | *C++17, QNX RTOS, OpenCV, dual Raspberry Pi, real-time systems* 2025

- Built a game you fight with your own shadow, with **under 80 ms** camera-to-screen latency, by writing a **C++17** vision and rendering system split across two Raspberry Pis on the **QNX real-time OS**.
- Kept traffic **under 12 kbps** at 30 fps, by designing a **~50-byte UDP protocol** with timestamp-matched buffering.

AI Air Hockey | *Computer vision, real-time control, OpenCV, embedded C* 2025

- Blocked human shots in **under 50 ms**, by tracking the puck in OpenCV, predicting wall bounces, and profiling the camera-to-motor loop.

TECHNICAL SKILLS

AI Systems & Inference Optimization: AI inference, INT8 quantization, pruning, TinyML, edge inference, TensorRT, kernel-level optimization, latency & power profiling, benchmarking, PyTorch, YOLOv8, physics-informed ML, sim-to-real

Signal Processing & Numerical Methods: Fourier & Laplace transforms, DFT/FFT, sampling, filter design, spectral analysis, sensor fusion, Kalman filtering, state estimation, robust regression, time-series analysis, MATLAB

Computer Architecture & Digital Design: SystemVerilog, RTL design, pipelining, hazard detection & forwarding, testbench verification, RISC-V, FPGA, Vivado, QuestaSim

Low-Level & Systems Programming: C/C++, Python, Rust, embedded C firmware, QNX RTOS, Linux, device drivers, I2C/SPI/UART, multithreading, POSIX sockets, CMake, Git, Docker, NumPy/SciPy, OpenCV, ROS 2